

Follow Your Nose: Experiencing Smell as a Design Material

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Figure 1: A visual depiction of a future world where there is a stronger awareness and appreciation for smell; reprint from [1].

Abstract

Over the past decades, olfactory interfaces have moved from the margins to the forefront of inquiry and design in HCI research and practice. Yet, while technical and interaction-focused work has advanced, there has been limited collective engagement with scent as a design material and its potential to shape multisensory experiences. Building on this growing interest, Follow Your Nose meet-up invites researchers, practitioners, and designers to collaboratively explore scent in this broader, practice-led context. We will draw on the olfactory design space through light, nose-first activities exploring how scents i) shape spatial experience, ii) evoke emotions and memories, iii) unfold over time, iv) reveal chemical contrasts that

influence perception, and v) convey recognisable qualities. Through these activities, we will reflect on scent as a medium for interaction and surface emerging questions, opportunities, and directions for research and practice in olfactory and multisensory HCI.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; • **Hardware** → **Emerging technologies**.

Keywords

Smell, Olfaction, Olfactory Interfaces, Smell Design, Smell Material, Digital Smell, Multisensory Experiences, Digital Scent Technology

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1 Introduction

1.1 Main Aim of the Meet-Up

Over the past decades, olfactory interfaces have moved from speculative experiments to a steadily growing area of HCI research and practice (e.g., [8], [16], [18], [20]). At CHI 2023, the panel “*Third Wave or Winter?*” [3] highlighted a crucial moment for the community: are we entering a sustainable “third wave” of digital smell, or facing another period of stagnation? That discussion emphasised both the exciting opportunities (e.g., new multisensory technologies and richer interaction possibilities) and the ongoing challenges (including safety and ethics, methods for evaluation, reproducibility, cultural variation, hardware robustness, and paths to commercialisation).

Since then, research and practice around olfactory interfaces have continued to grow rapidly both within the CHI community and in other related disciplines and venues [4]. Examples of this include scent-delivery device innovations [7], design futuring for smell [1], studies on the integration of scent and sound [19], and perceptual alignment between human-AI on smell experiences [23]. In addition, application examples include olfactory social robots [5], scent-mediated human-animal interaction systems [22], scent-enhanced meditation [11], and digital reading systems enhanced with olfactory for children [12]. The breadth and depth of research indicate a growing interest in elevating the sense of smell from the margins to the forefront of inquiry and design. This meet-up “*Follow your Nose*” builds directly on this momentum of the latest wave and aims to bring together researchers, practitioners, and designers to collectively explore how smell can be treated not only as an interface modality but as a design material in its own right.

By framing smell as a design material within the broader multisensory experience research efforts [21], we can help ensure that this emerging wave becomes a lasting and integrated dimension of human-computer interaction design.

1.2 Brief overview of the activities

This 90-minute meetup will centre on the olfactory design space [13], which identifies four key dimensions: (i) spatial, (ii) emotional, (iii) temporal, and (iv) chemical. Building on this framework, we also consider (v) olfactory quality, drawing on work in olfactory perception [9, 10]. Attendees are invited to bring notable scents of their own, alongside curated scents we will provide. During the session, we will:

- Follow our noses: Explore smell as a **spatial** entity by navigating the meetup space guided by scent, prioritising smell over sight or sound.
- Feel through our noses: Share personal stories about the **emotions** and **memories** brought to mind by the scents explored in the meet-up.
- Live through our noses: Investigate the **temporal** dimension of smell by noticing how perception changes over time – through onset, duration, repetition, and habituation.

- Sniff out the chemistry: Discover the **chemical** aspects of scent-intensity, contrasts, and familiar or surprising associations without lab coats.
- Sense with our noses: Attend to olfactory **qualities** themselves (e.g., floral, fruity, woody, earthy, musky, citrus, and beyond) using smell kits as inspiration for naming, distinguishing, and establishing meaning associated with olfactory characters.

Together, these activities provide a holistic, reflective nose-first experience of smell as a design material, allowing participants to consider its integration within multisensory interaction, explore new possibilities for design and research, and contribute to the ongoing momentum of olfactory HCI.

1.3 Organisers bios

Marianna Obrist: She is Professor of Multisensory Interfaces at UCL Department of Computer Science, investigating touch, taste, and smell as interaction modalities in Human-Computer Interaction, applied to VR/AR, automotive, and health/wellbeing. She is also co-founder and Chief Scientific Officer (CSO) of Hynt Labs, pioneering scent technology for sleep. Author of 150+ publications and co-author of *Multisensory Experiences* (Oxford University Press, 2nd ed., 2025), she advances understanding of the senses in technology.

Ceylan Beşevli: She is a Design Researcher and Research Fellow in UCL Computer Science, working on smell for health and wellbeing. Her research explores digital smell training, focusing on motivations and long-term engagement to envision future directions for smell care. She also investigates emerging multisensory interfaces such as acoustophoresis, examining their design implications. She holds a PhD in Interaction Design from Koç University and has taught design thinking to executives and industry professionals, as well as developing a course on Play & Innovation.

Emanuela Maggioni: She is a leading expert in olfaction and sensory science with over 18 years research and consultancy experience, including working with Unilever, Benetton and IFF. She has over 50 scientific publications and a PhD in Psychology of Smell Perception and Emotions and is a Royal Academy of Engineering Enterprise Fellow. She is also the Co-Founder and CEO/CTO of Hynt Labs Limited, pioneering novel digital scent technology, specifically for improved quality of sleep. More at hynt.net

Mei Kei Lai: She is Dean & Associate professor at the Faculty of Art and Design in Macao Polytechnic University. Her research interest is about exploring the possibilities of smell as the medium to create interactive user experience. She has been combining art practice with technology in the experiments including art exhibition, gameplay, reading, family bonding, virtual environment, smell map, etc. Her works have been presented in HCI, SIGDOC, TEL, ISEA, etc. She also held workshops related to digital olfaction. She is the author of the monograph “*Sniff the Future: Digital Olfactory Arts in Daily Life*” and “*Oh Sniff the Whiff: Exploring the Smellscapes in Macao*”.

Carlos Velasco: He is an Associate Professor of Marketing at BI Norwegian Business School, where he co-founded the Centre for Multisensory Marketing. He holds a D.Phil. in Experimental

Psychology from Oxford University and has worked on postdoctoral and consulting projects across Europe, Asia, and the Americas. His research sits at the intersection of psychology, marketing, and human–computer interaction, focusing on the principles and applications of multisensory experiences. He co-wrote the book "Multisensory experiences: Where the senses meet technology" (Oxford University Press) and "Digital dining: New innovations in food and technology" (Springer Nature). Carlos has also collaborated with global companies on multisensory design. More at carlosvelasco.info

1.4 Description of the communities of interest

The *Follow Your Nose* meet-up brings together communities invested in the design, engineering, psychology, and social science of multisensory experiences, where smell is both a perceptual and material element. For the HCI and multisensory interaction community, smell is an emerging design material for interactive systems, relevant to researchers exploring its integration with vision, sound, and touch. Sensory scientists, marketers, and service designers also engage with smell for its deep ties to perception, memory, and emotion. Specialised HCI communities further broaden this scope: child–computer interaction researchers study smell for play and learning [14]; human–food interaction researchers centre it in flavour and eating experiences [15]; and health researchers highlight its therapeutic and diagnostic potential, from olfactory rehabilitation [2] to early biomarkers of disease [6]. These communities demonstrate the social, scientific, and technological value of smell, situating the meet-up within the growing field of multisensory HCI [17], a space the organisers are shaping through their book *Multisensory Experiences: Where the Senses Meet Technology* [21].

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